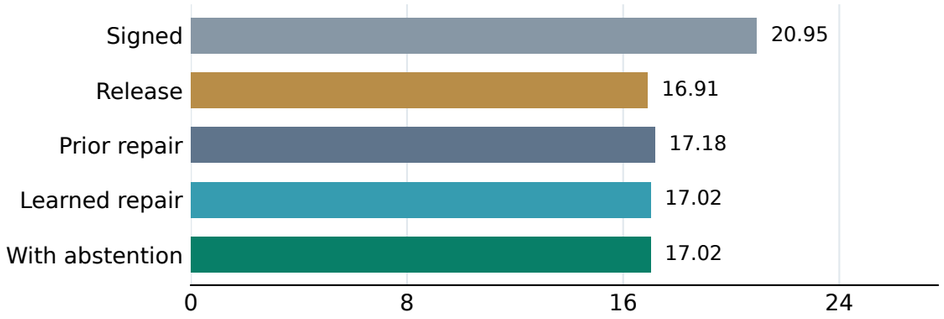
Coverage-based abstention: fresh budget validation

Fixed saved forests · zero new fits · training-supported structural cells · predictions frozen before labels

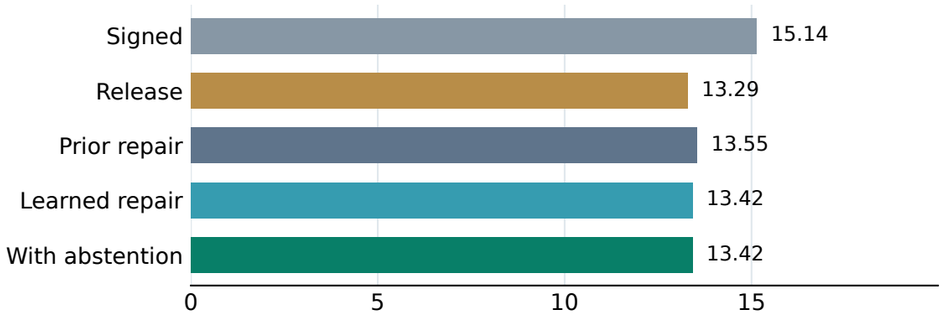
Evaluated policy: With abstention · Not eligible for promotion

General cohort · 32 fixed circuits

Policy MAE · LUT4 ↓



Policy positive-target MAPE · % ↓



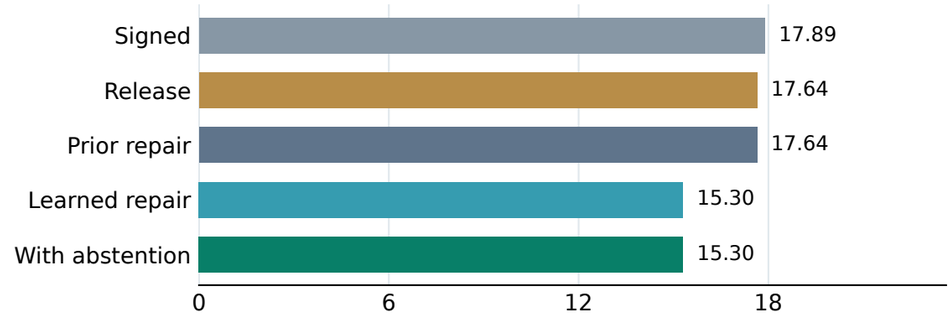
False accepts / false rejects · each budget assessed separately

Cap	Signed	Release	Prior	Learned	Abstain
32	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0
64	3 / 0	0 / 0	1 / 0	1 / 0	1 / 0
128	6 / 0	2 / 6	2 / 5	2 / 4	2 / 4

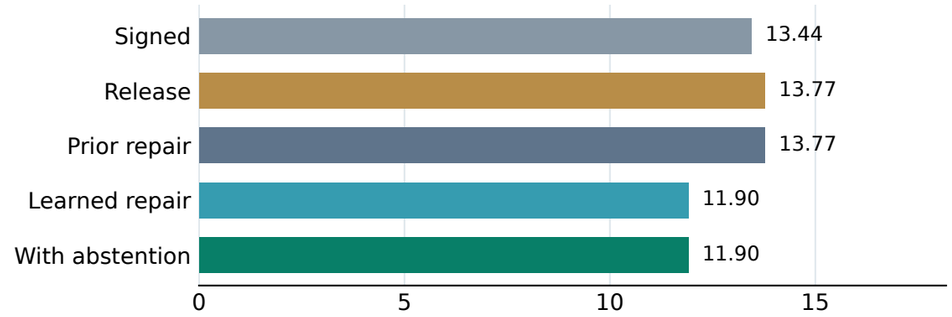
Feasible / infeasible: 32: 0/32 · 64: 4/28 · 128: 18/14

Conditional cohort · 6 active + 8 shared controls

Policy MAE · LUT4 ↓



Policy positive-target MAPE · % ↓



False accepts / false rejects · each budget assessed separately

Cap	Signed	Release	Prior	Learned	Abstain
32	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0
64	1 / 0	0 / 0	0 / 0	0 / 0	0 / 0
128	3 / 0	2 / 6	2 / 6	2 / 0	2 / 0

Feasible / infeasible: 32: 0/14 · 64: 2/12 · 128: 11/3

Unsupported: learned → abstention · n=8 · MAE: 27.32 → 26.71 LUT4 · FA/FR at 128: 1/0 → 0/0 · separate evaluation, never pooled

Unsupported panel fails release comparison: MAE 26.14 → 26.71 LUT4; false accepts at 64: 1 → 2. These failures remain in the eligibility decision.

Active-only and fixed unsupported-case results are reported separately; every predeclared gate is required for eligibility.

Eight controls are shared between cohorts. Conditional cases are selected by prediction disagreement; the cohorts are not pooled.

Measured target: isolated iCE40 LUT4 count. This is not a CPU speedup, application-transfer or universal hardware-safety result.